

[This question paper contains 12 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1755

G

Unique Paper Code : 2342571101

Name of the Paper : Programming Fundamentals
Using C++

Name of the Course : **B.Sc. (Prog.)**

Semester : I

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. **Section A** is compulsory.
3. Attempt any **four** questions from **Section B**.
4. Parts of a question must be answered together.

Section A

1. (a) (i) What is pointer variable? (3)
(ii) Consider the declarations:

```
int i = 10;  
int *j = &i;
```

P.T.O.

What is the difference between the operators * and & in the context of pointers?

(b) What is an inline function? What are the advantages of declaring functions as inline? (3)

(c) What would be output produced on the execution of following C++ code? (3)

```
#include<iostream>
using namespace std;
int main ()
{
    int x = 5, y = 5, z;
    x = ++x;
    y = --y;
    z = (x<<2) + (y>>2);
    cout<<x<<y<<z;
    return 0;
}
```

(d) (i) What is the difference between call by value and call by reference? (3)

(ii) Write a function that changes takes an input parameter and changes its value by incrementing it by 10. The changed value must be reflected in the main function.

(e) What would be output produced on the execution of following C++ code segments? (3)

(i)

```
int x = 1;
switch (x)
{
```

```
    case 1:    cout<< "Monday";
               case 2:    cout<< "Tuesday";
                           break;
    default:   cout<<"Weekend";
}
```

(ii)

```
int n, k = 5;
for(n=0 ; n<4 ; n++) ;
cout<<"\nn = "<<n<<" k = "<<k;
```

(iii)

```
int x = 12;
while(x>7)
{
    cout<<x<<endl;
    x -= 2;
}
```

(f) Write a function to swap the value of two numbers using pointers. (3)

(g) Identify the errors in the following C++ statements: (3)

(i) long float x = 3.5;

(ii) int code = three;

(iii) `int b(6);`

- (h) Rewrite the following code using switch case construct :
(3)

```
int number;
cout<<"Enter an integer: ";
cin>>number;
if (number == 1 || number == 2)
{
    cout<< "One or Two: "<< number<<endl;
}
else if (number == 3)
{
    cout<<"Three:"<<number<<endl;
}
else
    cout<<"Some other number";
```

- (i) Write a program to check if a given string is palindrome or not.

(Palindrome is a string that reads same forward as well as backwards, for example - MADAM)
(3)

- (j) Give the output produced upon execution of the following statements :
(3)

(i) `cout<<(21<<2);`

(ii) `cout<<(21^15);`

(iii) `cout<<(5+9*8/4);`

Section - B

2. (a) Consider the following code :
(5)

```
#include<iostream>
using namespace std ;
int main()
{
    int a,b;
    cout<<"\nEnter two numbers: ";
    cin>>a>>b;
    int x = a-b;
    try
    {
        if (x != 0)
            cout<<"\n"<<a/x;
        else
            throw(x);
    }
    catch(int i)
    {
        cout<<"\nException-Division by zero";
    }
    cout<<"\nEND";
}
```

What will be the output if

- (i) `a = 20, b = 15?`
(ii) `a = 10, b = 10?`
(b) (i) What is the purpose declaring data members of a class as static?
(ii) How are static variables of a class initialized?
(5)

- (iii) Declare a class having a static variable.
- (iv) Show how this variable will be accessed in the main() function.

- (c) What would be the output produced on the execution of the following code segment: (5)

```
int x = 0, i;
int a[] = {1, 4, 6, 8};
int *p = a;
for(i = 0; i < 5; i++)
{
    x = 2 * x + i;
}
cout << "Final value of x = "<<x<< endl;

cout<<*p;
*p++ ;

cout<<*p;
```

3. (a) What would be the output produced on the execution of the following code segment: (5)

```
int list[] = {1, 2, -5, -3, 6, 7, 0, -1, 6};
bool pos = true;
int i = 0;

while (pos && list[i] != 0 )
{
    pos = (list[i] > 0);
    i++;
}
cout<<"Final output: "<<pos<<endl;
cout<<"Value of i = "<<i;
```

- (b) Consider the following class definition: (5)

```
class circle
{
    float radius;
};
```

Define the following member functions:

- A default constructor that initializes radius to zero.
- A parameterized constructor that assigns the input value sent to radius.
- A function areaofcircle() for computing the area of a circle.

Create an object of class circle and call the above-mentioned functions in main() function.

- (c) Can constructors be overloaded? Write four important properties of constructor functions. (5)

4. (a) Identify errors in the following code segments, if any. Correct the code and give output produced upon the execution of the corrected code

- (i)

```
void carg(int *x, int *y)
{
    cout<<"Values are "<<x<<y<<endl;
    x = x+10;
    y = y+10;
}

int main()
{
    int x = 10;
    int y = 20;
    carg(x, y);
    cerr<<"After function call"<<x<<y;
}
```
- (ii)

```
void A(int a=9, int b)
{
    cout<<a<<and<<b<<end;
}

int main()
{
    A(10);
    return 0;
}
```

(b) Write a program that does the following: (5)

- (i) Creates a file File1 and write the following lines to it:
 "The biggest adventure is to live a life of your dreams."
- (ii) Count the number of characters in File1 and print this value.

- (c) (i) What is an exception? What are different types of exceptions in C++?
- (ii) Which type of exceptions are handled by a program? (5)

5. (a) Write a program to find the Least Common Multiple (L.C.D) of two numbers, say, a and b, taken as input from the user. (5)
- (b) Write a program to print the following pattern taking number of lines as input from the user: (5)

```

      *
    * * *
  * * * * *
* * * * * *
```

- (c) Consider the following class definition: (5)

```
class base
{
    int p1;
    protected:
        int p2;
    public:
        int p3;
};
```

What will be the visibility mode of p1, p2, p3 in 7. class derived in the following cases :

- (i) class derived: private base
{ };
- (ii) class derived: protected base
{ };
- (iii) can p1 be accessed in derived class?

6. (a) Write a program that defines a function template that does the following : (5)

- (i) Accepts one int and one char type argument
- (ii) Accepts one int and one float type argument
- (iii) Accepts two float type arguments

(b) Write a program to read an array of n elements, taking n as input from the user. Multiply every element of the array by 10 and print the resultant array. (5)

(c) What is polymorphism? Explain early binding and late binding in the context of polymorphism. (5)

(a) What is the use of following pointers? (5)

(i) this pointer

(ii) void pointer

(b) Consider the following program segments : (5)

```
class BC
{
    public: void show()
    {
        cout<< "I am in base class \n";
    }
};
class DC : public BC
{
    public: void show()
    {
        cout<< "I am in derived class\n";
    }
};
int main()
{
    BC *ptr;
    DC obj;
    ptr = &obj;
    ptr->show();
    return 0;
}
```

(i) What would be output produced upon the execution of the above code?

- (ii) What would you change in the above program to call the show() method of the derived class instead of the base class?
- (c) Write a function in C++ that takes a number 'n' as input and returns its factorial if 'n' is an odd, else the function should return 0. (5)